Smaran Rangarajan Bharadwaj

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EDUCATION

- **RV University, Bengaluru** · B.Tech (Hons.), Computer Science and Engineering — *Graduating: Aug 2026 · **CGPA**: 9.80*
- **Sri Kumaran Children's Home, Bengaluru** · Class 12 (CBSE) - PCMC — *2022 · Score: 95.2%*

TECHNICAL SKILLS

Languages: Python3, Go, C, C++, Java, JavaScript, YAML
Cloud & DevOps: AWS, Docker, Kubernetes, Git, GCP, Jira, Confluence
Databases: ClickHouse, PostgreSQL, MySQL, MongoDB, SQLite
Frameworks: PyTorch, TensorFlow, Node.js, Flask, Streamlit, gRPC, RESTful APIs, Ollama, Kafka, Flink, Unsloth, LangChain
Relevant Coursework: AWS Networking Masterclass, Big Data Computing, Distributed Systems

EXPERIENCE

Software Development Intern — *Jan 2026 - Jul 2026*
Aviatrix Systems Pvt. Ltd. — *Bengaluru*

- Architected and shipped a LLM-powered distributed multi-cloud (AWS/GCP) ETL pipeline leveraging Flink and ClickHouse analyzing risks across 22 resource types in under 100 seconds.
- Designed and shipped the scalable, cloud-native Data-Bridge backend in Go employing data modelingusing gRPC and ClickHouse materialized views, enabling sub-second queries across 10M+ records.
- Automated the CI/CD pipeline with GitHub Actions, reducing release time for multi-hour builds and deployments by 68%.

ML Project Intern — *Aug 2025 - Jan 2026*
Aeronautical Development Establishment (DRDO) — *Bengaluru*

- Architected a multi-stage mission debriefing pipeline processing 10,000+ flight parameters across 6 missions in under 15 minutes with parallelized processing.
- Engineered an event detection engine using Allen's Interval Algebra and expert systems logic to classify overlapping continuous and discrete parameter anomalies.

ML Engineer Intern — *Jun 2025 - Aug 2025*
GovernAIce — *Berkeley (Remote)*

- Designed an end-to-end multi-lingual RAG pipeline reducing compliance report generation time by 73% leveraging DeepSeek-R1:70B and MongoDB Atlas Vector Search.
- Developed a scalable batch ingestion pipeline processing 12,000+ unstructured legal records across 6 languages with XLM-RoBERTa vector embedding generation.

LEADERSHIP AND AWARDS

- Founded and established RV University's IEEE Student Branch, growing it from 5 founding ExeCom members to 60+ members across 3 society chapters and 20+ events.
- Awarded merit scholarships for 3 consecutive years, ranking 2nd among the university's top academic performers.
- Won 2nd place among 117 teams at the HPCC Safe Heavens Hackathon for developing a distributed computing solution for refugee support systems.
- Co-designed and published a patent for Gelos, a fitness-focused interactive game, alongside Dr. Phani Kumar Pullela.

PROJECTS

MITM Attack Simulator — *Network Security, Flask, PostgreSQL*

- Engineered a high-throughput network security toolkit processing 15,000+ packets per second, using Scapy, Kali Linux and Flask to simulate MITM attacks across a multi-VM bridged network.
- Architected a PostgreSQL relational schema to persist network logs and attacks with support for multi-user session audit queries.

ESP32 Flow Meter Fault Detection — *FreeRTOS, Mosquitto (MQTT), TensorFlow Lite*

- Engineered an IoT fault diagnosis system on ESP32 (FreeRTOS), deploying an optimized TensorFlow Lite model to classify flow meter faults into 4 states with 91.67% accuracy at 300–400ms inference latency.

Obstacle Avoidance for Visually Impaired — *Computer Vision, YOLO, Transfer Learning*

- Led a four-member team to develop a YOLOv8n-based obstacle detection system for the visually impaired, achieving 82.4% mAP on a self collected dataset with 132ms average inference latency per frame.